



- Q1-** On the basis of the following observations made with aqueous solutions, assign secondary valences to metals in the following compounds:

Formula	Moles of AgCl precipitated per mole of the compounds with excess AgNO_3
(i) $\text{PdCl}_2 \cdot 4\text{NH}_3$	2
(ii) $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$	2
(iii) $\text{PtCl}_4 \cdot 2\text{HCl}$	0
(iv) $\text{CoCl}_3 \cdot 4\text{NH}_3$	1
(v) $\text{PtCl}_2 \cdot 2\text{NH}_3$	0

- Q2-** Write the formulas for the following coordination compounds:

- (a) Tetraammineaquachloridocobalt(III) chloride
- (b) Potassium tetrahydroxidozincate(II)
- (c) Potassium trioxalatoaluminate(III)
- (d) Dichloridobis(ethane-1,2-diamine)cobalt(III)

- Q3-** Write the IUPAC names of the following coordination compounds:

- (a) $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$
- (b) $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$
- (c) $[\text{CoCl}_2(\text{en})_2]\text{Cl}$
- (d) $[\text{Co}(\text{NH}_3)_5(\text{CO}_3)]\text{Cl}$
- (e) $\text{Hg}[\text{Co}(\text{SCN})_4]$

- Q4-** Write the formulas for the following coordination compounds:

- (i) Tetraamminediaquacobalt(III) chloride
- (ii) Potassium tetracyanonickelate(II)
- (iii) Tris(ethane-1,2-diamine) chromium(III) chloride
- (iv) Amminebromidochloridonitrito-N-platinate(II)
- (v) Dichloridobis(ethane-1,2-diamine)platinum(IV) nitrate
- (vi) Iron(III) hexacyanidoferrate(II)

- Q5-** Write the IUPAC names of the following coordination compounds:

- (i) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
- (ii) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
- (iii) $\text{K}_3[\text{Fe}(\text{CN})_6]$
- (iv) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$
- (v) $\text{K}_2[\text{PdCl}_4]$
- (vi) $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NH}_2\text{CH}_3)]\text{Cl}$



- Q6-** Why is geometrical isomerism not possible in tetrahedral complexes having two different types of unidentate ligands coordinated with the central metal ion ?
- Q7-** Draw structures of geometrical isomers of $[\text{Fe}(\text{NH}_3)_2(\text{CN})_4]^-$
- Q8-** Out of the following two coordination entities which is chiral (optically active)? (a) $\text{cis}-[\text{CrCl}_2(\text{ox})_2]^{3-}$ (b) $\text{trans}-[\text{CrCl}_2(\text{ox})_2]^{3-}$
- Q9-** Indicate the types of isomerism exhibited by the following complexes and draw the structures for these isomers:
- (i) $\text{K}[\text{Cr}(\text{H}_2\text{O})_2(\text{C}_2\text{O}_4)_2]$
 - (ii) $[\text{Co}(\text{en})_3]\text{Cl}_3$
 - (iii) $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)](\text{NO}_3)_2$
 - (iv) $[\text{Pt}(\text{NH}_3)(\text{H}_2\text{O})\text{Cl}_2]$
- Q10-** Give evidence that $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Cl}$ are ionisation isomers.
- Q11-** The spin only magnetic moment of $[\text{MnBr}_4]^{2-}$ is 5.9 BM. Predict the geometry of the complex ion ?
- Q12-** Explain on the basis of valence bond theory that $[\text{Ni}(\text{CN})_4]^{2-}$ ion with square planar structure is diamagnetic and the $[\text{NiCl}_4]^{2-}$ ion with tetrahedral geometry is paramagnetic.
- Q13-** $[\text{NiCl}_4]^{2-}$ is paramagnetic while $[\text{Ni}(\text{CO})_4]$ is diamagnetic though both are tetrahedral. Why?
- Q14-** $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is strongly paramagnetic whereas $[\text{Fe}(\text{CN})_6]^{3-}$ is weakly paramagnetic. Explain.
- Q15-** Explain $[\text{Co}(\text{NH}_3)_6]^{3+}$ is an inner orbital complex whereas $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is an outer orbital complex.
- Q16-** Predict the number of unpaired electrons in the square planar $[\text{Pt}(\text{CN})_4]^{2-}$ ion.
- Q17-** The hexaquo manganese(II) ion contains five unpaired electrons, while the hexacyanoion contains only one unpaired electron. Explain using Crystal Field Theory.
- Q18-** Explain the bonding in coordination compounds in terms of Werner's postulates.
- Q19-** FeSO_4 solution mixed with $(\text{NH}_4)_2\text{SO}_4$ solution in 1:1 molar ratio gives the test of Fe^{2+} ion but CuSO_4 solution mixed with aqueous ammonia in 1:4 molar ratio does not give the test of Cu^{2+} ion. Explain why?
- Q20-** Explain with two examples each of the following: coordination entity, ligand, coordination number, coordination polyhedron, homoleptic and heteroleptic.
- Q21-** What is meant by unidentate, didentate and ambidentate ligands? Give two examples for each.
- Q22-** Specify the oxidation numbers of the metals in the following coordination entities:
- (i) $[\text{Co}(\text{H}_2\text{O})(\text{CN})(\text{en})_2]^{2+}$
 - (ii) $[\text{CoBr}_2(\text{en})_2]^+$
 - (iii) $[\text{PtCl}_4]^{2-}$
 - (iv) $\text{K}_3[\text{Fe}(\text{CN})_6]$
 - (v) $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$
- Q23-** Using IUPAC norms write the formulas for the following:
- (i) Tetrahydroxidozincate(II)
 - (ii) Potassium tetrachloridopalladate(II)
 - (iii) Diamminedichloridoplatinum(II)
 - (iv) Potassium tetracyanonickelate(II)
 - (v) Pentaamminenitrito-O-cobalt(III)



- (vi) Hexaamminecobalt(III) sulphate
- (vii) Potassium tri(oxalato)chromate(III)
- (viii) Hexaammineplatinum(IV)
- (ix) Tetrabromidocuprate(II)
- (x) Pentaamminenitrito-N-cobalt(III)

Q24- Using IUPAC norms write the systematic names of the following:

- (i) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
- (ii) $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NH}_2\text{CH}_3)]\text{Cl}$
- (iii) $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$
- (iv) $[\text{Co}(\text{NH}_3)_4\text{Cl}(\text{NO}_2)]\text{Cl}$
- (v) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$
- (vi) $[\text{NiCl}_4]^{2-}$
- (vii) $[\text{Ni}(\text{NH}_3)_6]\text{Cl}_2$
- (viii) $[\text{Co}(\text{en})_3]^{3+}$
- (ix) $[\text{Ni}(\text{CO})_4]$

Q25- List various types of isomerism possible for coordination compounds, giving an example of each.

Q26- How many geometrical isomers are possible in the following coordination entities?

- (i) $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$ (ii) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$

Q27- Draw the structures of optical isomers of:

- (i) $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$ (ii) $[\text{PtCl}_2(\text{en})_2]^{2+}$ (iii) $[\text{Cr}(\text{NH}_3)_2\text{Cl}_2(\text{en})]^+$

Q28- Draw all the isomers (geometrical and optical) of: (i) $[\text{CoCl}_2(\text{en})_2]^+$ (ii) $[\text{Co}(\text{NH}_3)\text{Cl}(\text{en})_2]^{2+}$ (iii) $[\text{Co}(\text{NH}_3)_2\text{Cl}_2(\text{en})]^+$

Q29- Write all the geometrical isomers of $[\text{Pt}(\text{NH}_3)(\text{Br})(\text{Cl})(\text{py})]$ and how many of these will exhibit optical isomers?

Q30- Aqueous copper sulphate solution (blue in colour) gives: (i) a green precipitate with aqueous potassium fluoride and (ii) a bright green solution with aqueous potassium chloride. Explain these experimental results.

Q31- What is the coordination entity formed when excess of aqueous KCN is added to an aqueous solution of copper sulphate? Why is it that no precipitate of copper sulphide is obtained when $\text{H}_2\text{S}(\text{g})$ is passed through this solution?

Q32- Discuss the nature of bonding in the following coordination entities on the basis of valence bond theory:

- (i) $[\text{Fe}(\text{CN})_6]^{4-}$ (ii) $[\text{FeF}_6]^{3-}$ (iii) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ (iv) $[\text{CoF}_6]^{3-}$

Q33- Draw figure to show the splitting of d orbitals in an octahedral crystal field.

Q34- What is spectrochemical series? Explain the difference between a weak field ligand and a strong field ligand.

Q35- What is crystal field splitting energy? How does the magnitude of Δ_o decide the actual configuration of d orbitals in a coordination entity?

Q36- $[\text{Cr}(\text{NH}_3)_6]^{3+}$ is paramagnetic while $[\text{Ni}(\text{CN})_4]^{2-}$ is diamagnetic. Explain why?

Q37- A solution of $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ is green but a solution of $[\text{Ni}(\text{CN})_4]^{2-}$ is colourless. Explain.



- Q38-** $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ are of different colours in dilute solutions. Why?
- Q39-** Discuss the nature of bonding in metal carbonyls.
- Q40-** Give the oxidation state, d orbital occupation and coordination number of the central metal ion in the following complexes:
- (i) $\text{K}_3[\text{Co}(\text{C}_2\text{O}_4)_3]$
 - (ii) $\text{cis-}[\text{CrCl}_2(\text{en})_2]\text{Cl}$
 - (iii) $(\text{NH}_4)_2[\text{CoF}_4]$
 - (iv) $[\text{Mn}(\text{H}_2\text{O})_6]\text{SO}_4$
- Q41-** Write down the IUPAC name for each of the following complexes and indicate the oxidation state, electronic configuration and coordination number. Also give stereochemistry and magnetic moment of the complex:
- (i) $\text{K}[\text{Cr}(\text{H}_2\text{O})_2(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$
 - (ii) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 - (iii) $[\text{CrCl}_3(\text{py})_3]$
 - (iv) $\text{Cs}[\text{FeCl}_4]$
 - (v) $\text{K}_4[\text{Mn}(\text{CN})_6]$
- Q42-** Explain the violet colour of the complex $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ on the basis of crystal field theory.
- Q43-** What is meant by the chelate effect? Give an example.
- Q44-** Discuss briefly giving an example in each case the role of coordination compounds in:
- (i) biological systems
 - (ii) medicinal chemistry
 - (iii) analytical chemistry
 - (iv) extraction/metallurgy of metals.
- Q45-** How many ions are produced from the complex $\text{Co}(\text{NH}_3)_6\text{Cl}_2$ in solution?
- (i) 6 (ii) 4 (iii) 3 (iv) 2
- Q46-** Amongst the following ions which one has the highest magnetic moment value?
- (i) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (ii) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (iii) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$
- Q47-** Amongst the following, the most stable complex is
- (i) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (ii) $[\text{Fe}(\text{NH}_3)_6]^{3+}$ (iii) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ (iv) $[\text{FeCl}_6]^{3-}$
- Q48-** What will be the correct order for the wavelengths of absorption in the visible region for the following: $[\text{Ni}(\text{NO}_2)_6]^{4-}$, $[\text{Ni}(\text{NH}_3)_6]^{2+}$, $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$?